

Vibronic Basis Formalism of Molecular Hamiltonians

” *Essentially, all models are wrong, but
some are useful.*

— **George E. P. Box**
Statistician

1.1 Vibronic basis representation

The interaction between electronic and nuclear degrees of freedom forms the foundation of molecular spectroscopy and non-adiabatic quantum dynamics. While the Born-Oppenheimer approximation provides a convenient separation between electronic and nuclear motion, many physically important processes such as vibronic coupling, photoionization, and electronic population transfer require a framework in which both contributions are treated simultaneously.

In this project, the molecular Hamiltonian and dipole operators are expressed in a vibronic basis constructed from products of electronic parent-ion states and their associated vibrational eigenstates. This representation naturally leads to a block-structured Hamiltonian, where diagonal blocks describe vibrational ladders on individual electronic surfaces, while off-diagonal blocks account for interstate couplings weighted by Franck–Condon overlaps.

The resulting formalism provides a computationally efficient and physically transparent framework for constructing large vibronic Hamiltonians from smaller electronic building blocks. Furthermore, the factorization of coupling and transition matrix elements into electronic and vibrational contributions enables the direct incorporation of Franck-Condon factors into both Hamiltonian assembly and transition dipole calculations.

1.1.1 Hamiltonian in the vibronic basis

The total molecular Hamiltonian under the Born-Oppenheimer Approximation can be expressed as:

$$\hat{H} = \hat{H}_{electronic} + \hat{H}_{nuclear} \quad (1.1)$$

The total wavefunction as a function of the electronic and the nuclear basis is, assuming multiple parent ions and each with its own set of vibrational states:

$$|\Psi_{Ii}\rangle = |X_I\rangle |\phi_i^I\rangle \quad (1.2)$$

where X_I is the electronic state of parent ion I and ϕ_i^I is the i^{th} vibrational state of parent ion I . Thus, the matrix element in this vibronic basis of general hamiltonian will be:

$$\hat{H}_{Ii,Jj} = \langle X_I \phi_i^I | \hat{H} | X_J \phi_j^J \rangle \quad (1.3)$$

$$H_{Ii,Jj} = \langle X_I \phi_i^I | \hat{H}_e | X_J \phi_j^J \rangle + \langle X_I \phi_i^I | \hat{H}_n | X_J \phi_j^J \rangle \quad (1.4)$$

The matrix elements naturally separate into two distinct cases: first, when there are two states from the same parent ion but different vibrational states and second, when the two states are from different parent ions to begin with.

Case 1: Hamiltonian for same Parent Ion

For same parent ion, $I = J$. Since the electronic states are orthonormal $\langle X_I | X_J \rangle = \delta_{IJ}$, which simplifies the Hamiltonian matrix elements to:

$$H_{Ii,Ij} = \langle \phi_i^I | H_n^I | \phi_j^I \rangle + \langle \phi_i^I | E_I(R) | \phi_j^I \rangle \quad (1.5)$$

Since the vibrational states are eigenstates of the nuclear Hamiltonian on potential energy surface of the parent ion I :

$$H_n^I \phi_i^I = E_{vib,i}^I \phi_i^I \quad (1.6)$$

Therefore, the matrix becomes diagonal in the vibrational states:

$$H_{Ii,Ij} = (E_I + E_{vib,i}^I) \delta_{ij} \quad (1.7)$$

Thus, for a single electronic state, the Hamiltonian takes the form of a vibrational ladder shifted by the electronic energy:

$$H = \begin{pmatrix} E_I + E_{vib,0} & 0 & 0 \\ 0 & E_I + E_{vib,1} & 0 \\ 0 & 0 & E_I + E_{vib,2} \end{pmatrix} \quad (1.8)$$

Case 2: Hamiltonian for different Parent Ions

For $I \neq J$, under the Born-Oppenheimer Approximation, we get:

$$\langle X_I | H_e | X_J \rangle = 0 \quad (1.9)$$

The vibronic coupling is expressed through the nuclear motion:

$$H_{Ii,Jj} = V_{IJ} \langle \phi_i^I | \phi_j^J \rangle \quad (1.10)$$

where the overlap $\langle \phi_i^I | \phi_j^J \rangle$ is exactly the Franck-Condon factor. Thus, the hamiltonian for states arising from different parent ions is simply:

$$H_{Ii,Jj} = V_{IJ} F_{ij} \quad (1.11)$$

where, $F_{ij} = \langle \phi_i^I | \phi_j^J \rangle$

Thus, the overlap hamiltonian in full-dimensionality is a block matrix of the form:

$$H = \begin{pmatrix} H^1 & V_{12}F & V_{13}F \\ V_{21}F & H^2 & V_{23}F \\ V_{31}F & V_{32}F & H^3 \end{pmatrix} \quad (1.12)$$

with

$$H_{Ii,Jj} = (E_I + E_{vib,i}^I) \delta_{IJ} \delta_{ij} + V_{IJ} F_{ij} (1 - \delta_{IJ}) \quad (1.13)$$

1.1.2 Dipole in the vibronic basis

With the same wavefunction form as before, the dipole operator ($\mu = -er$) matrix element as the following form:

$$\mu_{Ii,Jj} = \langle X_I \phi_i^I | r | X_J \phi_j^J \rangle \quad (1.14)$$

Separating the electronic and vibrational parts, gives us the following form:

$$\mu_{Ii,Jj} = \langle X_I | r | X_J \rangle \langle \phi_i^I | \phi_j^J \rangle \quad (1.15)$$

Defining the electronic transition dipole as:

$$\mu_{IJ} = \langle X_I | r | X_J \rangle \quad (1.16)$$

we get:

$$\mu_{Ii,Jj} = \mu_{IJ} F_{ij} \quad (1.17)$$

This leads to a more simplified result than the hamiltonian for the two cases:

Case 1: Dipole for same Parent Ion

$$\mu_{Ii,Jj} = \mu_{II} \delta_{ij} \quad (1.18)$$

which gives the physically understandable idea that the dipole is usually zero or small for transitions between the same parent ion unless there is a permanent dipole change.

Case 2: Dipole for different Parent Ions

$$\mu_{Ii,Jj} = \mu_{IJ} F_{ij} \quad (1.19)$$

in this case, the Franck-Condon factors weight the transitions.

1.1.3 Final form

The Hamiltonian matrix elements may therefore be written in a condensed form as:

$$H_{Ii,Jj} = (E_I + E_{vib,i}^I) \delta_{IJ} \delta_{ij} + V_{IJ} F_{ij} (1 - \delta_{ij}) \quad (1.20)$$

while the dipole operator takes the form:

$$\mu_{Ii,Jj} = \mu_{IJ} F_{ij}, \quad (1.21)$$

with $F_{ij} = \langle \phi_i^I | \phi_j^J \rangle$. Physically, the diagonal blocks correspond to vibrational ladders built upon each electronic surface, whereas the off-diagonal blocks describe electronic coupling weighted by Franck–Condon overlaps. Similarly, transition dipole matrix elements factorize into an electronic transition dipole moment multiplied by the corresponding vibrational overlap.